Physics

Answer Key

MECHANICS

1. $\dfrac{(\vec{r}_1 - \vec{r}_2)}{|\vec{r}_1 - \vec{r}_2|} = \dfrac{(\vec{v}_2 - \vec{v}_1)}{|\vec{v}_2 - \vec{v}_1|}$

2. $\dfrac{\tau_A}{\tau_B} = \dfrac{\eta}{\sqrt{\eta^2 - 1}} = 1.8$

3. $t = \dfrac{2a}{3v}$

4. It is seen from Fig. 1*a* that the points *A* and *B* converge with velocity $v - u\cos\alpha$, where the angle α varies with time. The points merge provided the following two conditions are met:

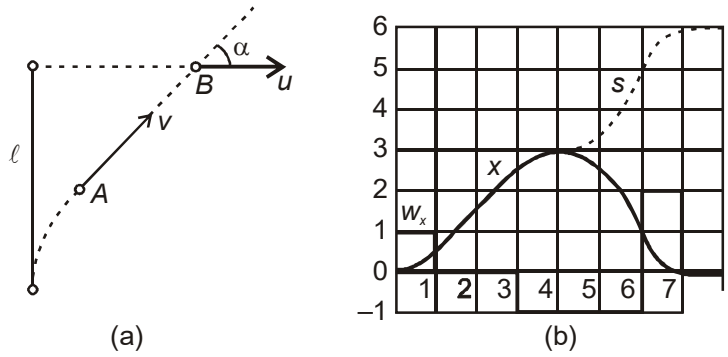


Fig. 1

$$\int_0^{\tau} (v - u\cos\alpha)\, dt = \ell, \qquad \int_0^{\tau} v\cos\alpha\, dt = u\tau,$$

where τ is the sought time. It follows from these two equations that

$$\tau = \frac{v\ell}{(v^2 - u^2)}$$

5. $CD = \dfrac{\ell}{\sqrt{\eta^2 - 1}}$

6. (a) $y = \left(\frac{b}{2a}\right)x^2$

(*b*) $R = \frac{v^2}{w_n} = \frac{v^2}{\sqrt{w^2 - w_\tau^2}} = \left(\frac{a}{b}\right)\left[1 + \left(\frac{xb}{a}\right)^2\right]^{3/2}$

7. (a) $w = 2av^2$, $R = \frac{1}{2}a$

(b) $w = \frac{bv^2}{a^2}$, $R = \frac{a^2}{b}$

8. (a) $y = \frac{v^2}{\beta x}$ (hyperbola)

(b) $y = \frac{\sqrt{2wx}}{\omega}$ (parabola)

9. (a) $\omega = at\sqrt{1 + \left(\frac{bt}{a}\right)^2} = 8 \text{ rad/s}$, $\beta = a\sqrt{1 + \left(\frac{2bt}{a}\right)^2} = 1.3 \text{ rad/s}^2$

(b) 17°

10. $\omega = \omega_0\sqrt{1 + \left(\frac{\beta_0 t}{\omega_0}\right)^2} = 0.6 \text{ rad/s}$, $\beta = \beta_0\sqrt{1 + \omega_0^2 t^2} = 0.2 \text{ rad/s}^2$

11. When $t \le t_0$, the accelerations $w_1 = w_2 = \frac{at}{(m_1 + m_2)}$; when $t \ge t_0$

$w_1 = \frac{kgm_2}{m_1}$, $w_2 = \frac{(at - km_2 g)}{m_2}$. Here $t_0 = kgm_2\frac{(m_1 + m_2)}{am}$. See Fig. 4.

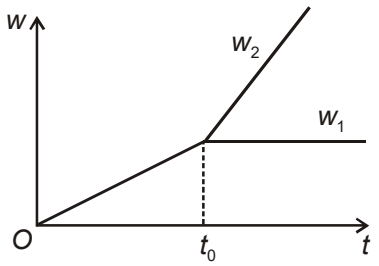


Fig. 2

12. $w_1 = \frac{4m_1m_2 + m_0(m_1 - m_2)}{4m_1m_2 + m_0(m_1 + m_2)}g.$

13. $w_{\min} = g\frac{(1-k)}{(1+k)}.$

14. $w = \dfrac{mg\sin\alpha}{M + 2m\,(1-\cos\alpha)}$.

15. $\varepsilon = \dfrac{\ell}{\left(\dfrac{\chi}{m\omega^2 - 1}\right)}$. Is independent of the rotation direction.

16. $t = \dfrac{\pi}{\omega}$; $s = \dfrac{2F_0}{m\omega^2}$; $v_{max} = \dfrac{F_0}{m\omega}$.

17. $v = \dfrac{v_0}{(1+\cos\varphi)}$.

Instruction: Here $w_\tau = -w_x$, and therefore $v = -v_x$ + const. From the initial condition it follows that const. = v_0. Besides, $v_x = v\cos\varphi$.

18. For $n < 1$, including negative values.

19. $t = \dfrac{\ell_0^2}{2v_0R}$

20. $A = \dfrac{kmg\ell_0}{2}\dfrac{1-\cos\theta}{(\sin\theta + k\cos\theta)\cos\theta} = 0.09$ J

21. $r = \left(\dfrac{g}{\omega^2}\right)\tan\theta = 0.8$ cm, $T = \dfrac{mg}{\cos\theta} = 5$ N

22. (a) $\Delta\ell > \dfrac{3mg}{\chi}$;

(b) $h = \left(\dfrac{1+\chi\Delta\ell}{mg}\right)^2 \dfrac{mg}{8\chi} = \dfrac{8mg}{\chi}$

23. (a) $A = -\mu gh$, where $\mu = \dfrac{mM}{(m+M)}$;

(b) Yes

24. $\sin\theta_{max} = \dfrac{m_2}{m_1}$

25. Results to be proved.

26. $v = \sqrt{2gh\ln\left(\dfrac{\ell}{h}\right)}$

27. (a) Relative to all points of the straight line drawn at right angles to the wall through the point O;

(b) $|\Delta\vec{M}| = 2\,mv\ell\cos\alpha$.

28. $F = \dfrac{m\omega_0^2 r_0^4}{r^3}$

29. $r_{\min} = \left(\frac{\gamma m_s}{v_0^2}\right)\left[\sqrt{1+\left(\frac{\ell v_0^2}{\gamma m_s}\right)^2} - 1\right]$, where m_s is the mass of the Sun.

30. $\Delta t \approx \frac{2\pi}{\sqrt{\gamma M}} \frac{r^{3/2}}{\frac{3\Delta r}{2r} + \delta} = \begin{cases} 4.5 & \text{days} \quad (\delta = 0), \\ 0.84 & \text{hour} \quad (\delta = 2). \end{cases}$

31. The decrease in the total energy E of the satellite over the time interval dt is equal to $-dE = Fv\,dt$. Representing E and v as functions of the distance r between the satellite and the centre of the Moon, we can reduce this equation to the form convenient for integration. Finally, we get $\tau \approx \left(\sqrt{n} - 1\right)\frac{m}{\alpha\sqrt{gR}}$

32. $v_3 \approx \sqrt{2v_1^2 + \left(\sqrt{2} - 1\right)^2 V_1^2} \approx 17\ \text{km/s}$. Here $v_1^2 = \frac{\gamma M_E}{R}$, M_E and R are the mass and the radius of the Earth; $V_1^2 = \frac{\gamma M_s}{r}$, M_s is the mass of the Sun, r is the radius of the Earth's orbit.

33. $\omega = \sqrt{6F\frac{\sin\varphi}{m\ell}}$

34. $n = (1 + k^2)\ \frac{\omega_0^2 R}{8\pi k\,(k+1)\,g}$.

35. $F_{\max} = \frac{3kmg}{(2-3k)}$; $w_{\max} = \frac{2kg}{(2-3k)}$

36. (a) $w_x = \frac{F\left(\cos\alpha - \frac{r}{R}\right)}{m(1+\gamma)}$;

(b) $A = \frac{F^2 t^2\left(\cos\alpha - \frac{r}{R}\right)^2}{2m(1+\gamma)}$

37. $w = \frac{3g(M+3m)}{\left(M + 9m + \frac{I}{R^2}\right)}$

38. $v_0 = \sqrt{\frac{1}{3}gR\,(7\cos\alpha - 4)} = 1.0\ \text{m/s}$

39. $T = \frac{7}{10}mv^2\left(1 + \frac{\frac{2}{7}r^2}{R^2}\right)$

40. $N = \frac{1}{24}m\omega^2\ell^2\sin 2\theta$

41. (a) $\varphi = \left(\frac{\ell}{2\pi r^3 \Delta r G}\right)\cdot N$;

(b) $\varphi = \left(\frac{2\ell}{\pi r^4 G}\right)\cdot N$

42. $N = \pi\left(d_2^4 - d_1^4\right)\dfrac{G\varphi}{32\ell} = 0.5$ kN.m.

43. $p_1 > p_2$, $v_1 < v_2$. The density of streamlines grows on transition from point 1 to point 2.

44. $F = \dfrac{\rho g h (S - s)^2}{S} = 6$ N

45. (a) $\omega = \omega_2 \dfrac{R_1^2 R_2^2}{R_2^2 - R_1^2}\left(\dfrac{1}{R_1^2} - \dfrac{1}{r^2}\right)$;

(b) $N = 4\pi\eta\omega_2 \dfrac{R_1^2 R_2^2}{R_2^2 - R_1^2}$

46. $e^{\alpha\Delta x} = 5$.

THERMODYNAMICS

1. $\Delta T = (mg + p_0\ \Delta S)\dfrac{\ell}{R} = 0.9$ K

2. $p_{\min} = 2R\sqrt{\alpha T_0}$

3. (a) $p = p_0\ (1 - ah)^n, h < \dfrac{1}{a}$; (b) $p = \dfrac{p_0}{(1 + ah)^n}$. Here $n = \dfrac{Mg}{aRT_0}$.

4. $\chi = \dfrac{V^2 (V - b)^2}{\left[RTV^3 - 2a(V - b)^2\right]}$

5. $T > \dfrac{a}{bR}$

6. $Q = \nu R T_0\left(1 - \dfrac{1}{n}\right) = 2.5$ kJ

7. See Fig. 1 where V is an isochore, p is an isobaric line, T is an isothermal line, and S is an adiabatic line.

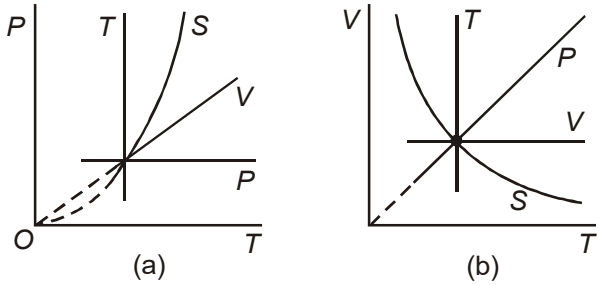


Fig. 1

8. $v = \sqrt{\dfrac{2\gamma RT}{(\gamma - 1)M}} = 3.3$ km/s

9. (a) $C = C_V + \frac{R}{\alpha V}$;

(b) $C = C_V + \frac{R}{(1+\alpha V)}$

10. (a) $A = \alpha \ln\eta - RT_0 \frac{(\eta - 1)}{(\gamma - 1)}$;

(b) $pV^{\gamma} e^{\alpha \frac{(\gamma-1)}{pV}} = \text{const.}$

11. $h = \frac{kT \ln\left(\frac{n_2}{n_1}\right)}{(m_2 - m_1)g}$

12. $\omega = \sqrt{\left(\frac{2RT}{M\ell^2}\right) \ln\eta} = 280 \text{ rad/s}$

13. $\eta = 1 - \frac{2T_3}{(T_1 + T_2)}$

14. In both cases $\eta = 1 - \frac{\ln n}{n-1}$

15. $\eta = \frac{(\tau - 1)\ln n}{\tau \ln n + (\tau - 1)\frac{\gamma}{(\gamma - 1)}}$

16. $\eta = 1 - 2\frac{\gamma + \sqrt{\tau}}{(1+\gamma)(1+\sqrt{\tau})}$

17. (a) $C = -\frac{\alpha}{T}$;

(b) $Q = \alpha \ln\left(\frac{T_1}{T_2}\right)$;

(c) $A = \alpha \ln\left(\frac{T_1}{T_2}\right) + C_V (T_1 - T_2)$

18. $\alpha = \left[\frac{\rho g h + p_0 h}{(\ell - h)}\right] \frac{d}{4\cos\theta}$

19. $h = \dfrac{2\alpha\cos\theta}{\rho g x \delta\varphi}$

20. $A' = F + pV\ln\left(\dfrac{p}{p_0}\right)$, where $F = 8\pi R^2\alpha$, $p = p_0 + \dfrac{4\alpha}{R}$, $V = \dfrac{4}{3}\pi R^3$.

ELECTRODYNAMICS

1. (a) $E = \dfrac{\lambda_0}{4\varepsilon_0 R}$;

   (b) $E = \dfrac{\lambda_0 R^2}{4\varepsilon_0\left(x^2 + R^2\right)^{3/2}}$.

   For $x >> R$ the strength $E \approx \dfrac{p}{4\pi\varepsilon_0 x^3}$, where $p = \pi R^2\lambda_0$.

2. $\vec{E} = -\dfrac{1}{3}\dfrac{\vec{k}\sigma_0}{\varepsilon_0}$ where $\vec{k}$ is the unit vector of the z axis with respect to which the angle θ is read off. Clearly, the field inside the given sphere is uniform.

3. $\varphi = \dfrac{\sigma\ell}{2\varepsilon_0}\left(\sqrt{1+\left(\dfrac{R}{\ell}\right)^2} - 1\right)$, $E = \dfrac{\sigma}{2\varepsilon_0}\left(1 - \dfrac{\ell}{\sqrt{\ell^2 + R^2}}\right)$. When $\ell \to 0$, then $\varphi = \dfrac{\sigma R}{2\varepsilon_0}, E = \dfrac{\sigma}{2\varepsilon_0}$; when $\ell >> R$, then $\varphi \approx \dfrac{q}{4\pi\varepsilon_0\ell}$, $E \approx \dfrac{q}{4\pi\varepsilon_0\ell^2}$, where $q = \sigma\pi R^2$.

4. $\vec{E} = -2(ax\hat{i} + ay\hat{j} + bz\hat{k})$, $E = 2\sqrt{a^2\left(x^2 + y^2\right) + b^2z^2}$.

   (a) An ellipsoid of revolution with semiaxes $\sqrt{\dfrac{\varphi}{a}}$ and $\sqrt{\dfrac{\varphi}{b}}$.

   (b) In the case of $\varphi > 0$, a single-cavity hyperboloid of revolution; when $\varphi = 0$, a right round cone; when $\varphi < 0$, a two-cavity hyperboloid of revolution.

5. $\varphi = ay\left(\dfrac{y^2}{3} - x^2\right) + \text{const}$.

6. $\rho = -6\varepsilon_0 a$

7. (a) $\sigma = \dfrac{\lambda}{2\pi\ell}$;

   (b) $\sigma(r) = \dfrac{\lambda}{2\pi\sqrt{\ell^2 + r^2}}$

8. Result to be proved.

9. (a) $C = \varepsilon_0\left(\varepsilon_2 - \varepsilon_1\right)\dfrac{S}{d\ln\left(\dfrac{\varepsilon_2}{\varepsilon_1}\right)}$;

   (b) $\rho' = \dfrac{-q\left(\varepsilon_2 - \varepsilon_1\right)}{dS\varepsilon^2}$

10. $V = R_1E_1\left[\ln\left(\frac{R_2}{R_1}\right) + \left(\frac{\varepsilon_1}{\varepsilon_2}\right)\ln\left(\frac{R_3}{R_2}\right)\right]$

11. $C \approx 2\pi\varepsilon_0\varepsilon a$.

Instruction: When $b >> a$, the charges can be assumed to be distributed practically uniformly over the surfaces of the balls.

12. $q_1 = \varepsilon C_2,\ q_2 = \frac{-\varepsilon C_1C_2}{(C_1 + C_2)}$

13. $W = -\frac{2\ln 2}{4\pi\varepsilon_0}\frac{q^2}{a}$

14. $Q = \frac{\varepsilon^2 CC_0}{(2C + C_0)}$.

15. $A = \left(\frac{q^2}{8\pi\varepsilon_0}\right)\left(\frac{1}{a} - \frac{1}{b}\right)$

16. $N = (\varepsilon - 1)\frac{\varepsilon_0 R^2V^2}{4d}$

17. (a) $\frac{5}{6}R$;

(b) $\frac{7}{12}R$;

(c) $\frac{3}{4}R$

18. Imagine the voltage V to be applied across the points A and B. Then $V = IR = I_0R_0$, where I is the current carried by the lead wires, I_0 is the current carried by the conductor AB.

The current I_0 can be represented as a superposition of two currents. If the current I flowed into point A and spread (all over) the infinite wire grid, the conductor AB would carry (because of symmetry) the current $\frac{I}{4}$. Similarly, if the current I flowed into the grid from infinity and left the grid through point B, the conductor AB would also carry the current $\frac{I}{4}$. Superposing both of these solutions, we obtain $I_0 = \frac{I}{2}$. Therefore, $R = \frac{R_0}{2}$.

19. $\rho = \frac{4\pi\Delta tab}{(b - a)C\ln\eta}$

20. Result to be proved.

21. $R = R_2 - R_1$, $\Delta\varphi = 0$ in the source of current with internal resistance R_2.

22. $I = \frac{(R_1\varepsilon_2 - R_2\varepsilon_1)}{(RR_1 + R_1R_2 + R_2R)} = 0.02$ A, the current is directed from the left to the right (see Fig. 2).

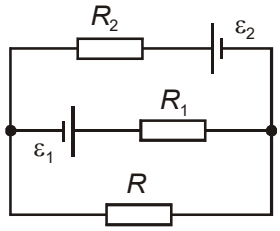


Fig. 2

23. $R_{AB} = \dfrac{r\,(r+3R)}{(R+3r)}$

24. $T - T_0 = \left(1 - e^{-\frac{kt}{C}}\right)\dfrac{V^2}{kR}$

25. (a) $q = q_0 e^{-\frac{t}{\varepsilon_0 \varepsilon \rho}}$;

(b) $Q = \left(\dfrac{1}{a} - \dfrac{1}{b}\right)\dfrac{q_0^2}{8\pi\varepsilon_0\varepsilon}$

26. (a) $\rho(x) = -\dfrac{4}{9}\,\varepsilon_0 a x^{-\frac{2}{3}}$

(b) $j = \dfrac{4}{9}\,\varepsilon_0 a^{\frac{3}{2}}\sqrt{\dfrac{2e}{m}}$

27. $B = n\mu_0 I\,\dfrac{\tan\left(\frac{\pi}{n}\right)}{2\pi R}$, for $n \to \infty$, $B = \dfrac{\mu_0 I}{2R}$

28. In the half-space with the straight wire, $B = \dfrac{\mu_0 I}{2\pi r}$, r is the distance from the wire. In the other half-space $B \equiv 0$.

29. (a) $F_1 = \dfrac{\mu_0 I^2}{4R} = 0.20$ mN/m;

(b) $F_1 = \dfrac{\mu_0 I^2}{\pi \ell} = 0.13$ mN/m

30. $B = \dfrac{\Delta m g \ell}{NIS} = 0.4\ T$

31. $F_1 = \dfrac{\mu_0}{4\pi}\,\dfrac{2I_1 I_2}{b}\,\ln\left(1 + \dfrac{b}{a}\right)$

32. $\Delta p = \dfrac{IB}{a} = 0.5$ kPa

33. $F = \left(\frac{\mu_0}{4\pi}\right) 6\pi R^2 I \frac{p_m x}{\left(R^2 + x^2\right)^{5/2}}$

34. $\varepsilon_i = \frac{\frac{1}{2}\left(\omega a^3 B^2 + 2mg \sin\omega t\right)}{aB}$

35. $q = \frac{\mu_0 aI}{2\pi R} \ln \frac{b+a}{b-a}$, *i.e.* is independent of L.

36. (a) $s = \frac{v_0 mR}{\ell^2 B^2}$,

(b) $Q = \frac{1}{2} m v_0^2$

37. $L = \frac{\mu_0}{2\pi} \mu N^2 a \ln\left(1 + \frac{a}{b}\right)$

38. $I = \frac{\pi aB}{\mu_0 \left(\ln \frac{8a}{b} - 2\right)} = 50 \text{ A}$

39. Result to be proved.

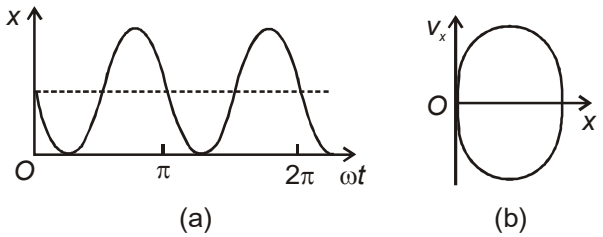
40. $v = \frac{V}{rB \ln\left(\frac{b}{a}\right)}$, $\frac{q}{m} = \frac{V}{r^2 B^2 \ln\left(\frac{b}{a}\right)}$

41. $v = \frac{1}{2}\frac{rqB}{m}$, $\rho = \frac{r}{2}$

OSCILLATIONS AND WAVES

1. (a) The amplitude is equal to $\frac{a}{2}$, and the period is $T = \frac{\pi}{\omega}$, see Fig. 1a;

(b) $v_x^2 = 4\omega^2 x (a - x)$, see Fig. 1b.



(a) (b)

Fig. 1a and b

2. $\frac{dP}{dx} = \frac{1}{\pi\sqrt{a^2 - x^2}}$

3. $T = 2\pi\sqrt{\dfrac{\eta\ell}{g(\eta - 1)}} = 1.1$ s

4. $T = 2\pi\sqrt{\dfrac{m}{S\rho g(1 + \cos\theta)}} = 0.8$ s

5. $T = 2\pi\sqrt{\dfrac{\ell}{|\vec{g} - \vec{w}|}} = 0.8$ s, where $|\vec{g} - \vec{w}| = \sqrt{g^2 + w^2 - 2gw\cos\beta}$

6. (a) $F = mg\left(1 + \dfrac{a\omega^2}{g}\cos\omega t\right)$, see Fig. 2;

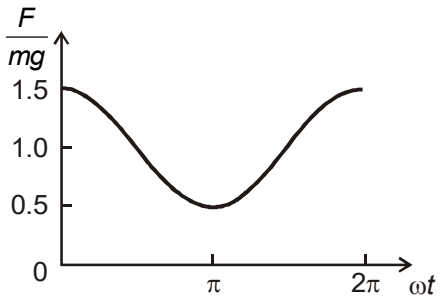


Fig. 2

(b) $a_{min} = \dfrac{g}{\omega^2} = 8$ cm;

(c) $a = \left(\omega\sqrt{\dfrac{2h}{g}} - 1\right)\dfrac{g}{\omega^2} = 20$ cm

7. Let us write the motion equation in projections on the *x* and *y* axes:

$\ddot{x} = \omega\dot{y}$, $\ddot{y} = -\omega x$, where $\omega = \dfrac{a}{m}$

8. (a) $T = 2\pi\sqrt{\dfrac{\ell}{3g}} = 1.1$ s;

(b) $E = \dfrac{1}{2}mg\ell\alpha^2 = 0.05$ J

9. $\langle T\rangle = \dfrac{1}{8}mg\ell\theta_0^2 + \dfrac{1}{12}m\ell^2\dot{\theta}_0^2$

10. $\omega_0 = \sqrt{\dfrac{2mg\cos\alpha}{MR + 2mR(1 + \sin\alpha)}}$

11. $\dfrac{\omega_2}{\omega_1} = \sqrt{1 + \dfrac{2m_O}{m_C}} \approx 1.9$, where m_O and m_C are the masses of oxygen and carbon atoms.

12. $x = (1 - \cos\omega t)\dfrac{g}{\omega^2}$, where $\omega = \dfrac{\ell B}{\sqrt{mL}}$

13. (a) $\dot{\varphi}(0) = -\beta\varphi_0$, $\ddot{\varphi}(0) = (\beta^2 - \omega^2)\varphi_0$;

(b) $t_n = \frac{1}{\omega}\left(arc\tan\frac{\omega^2 - \beta^2}{2\beta\omega} + n\pi\right)$, where n = 0, 1, 2

14. (a) $I = I_m \sin\omega_0 t$, where $I_m = V_m\sqrt{\frac{C}{L}}$, $\omega_0 = \frac{1}{\sqrt{LC}}$;

(b) $\xi_S = \frac{V_m}{\sqrt{2}}$.

15. (a) $t_n = \frac{\pi n}{\omega}$;

(b) $t_n = \frac{1}{\omega}\left[arc\tan\left(-\frac{\beta}{\omega}\right) + \pi n\right]$. Here n = 0, 1, 2, . . .

16. $I = I_m \cos(\omega t + \varphi)$, where $I_m = \frac{V_m}{R}\sqrt{1 + (\omega RC)^2}$ and $\tan\varphi = \omega RC$.

17. $P_2 = \frac{\frac{1}{2}(V^2 - V_1^2 - V_2^2)}{R} = 30$ W

18. Result to be proved.

19. Results to be proved.

20. See Fig. 3.

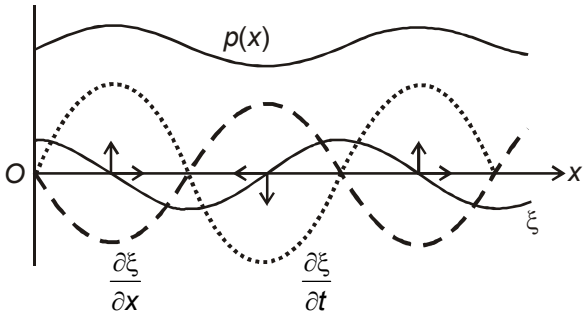


Fig. 3

21. See Fig. 4, for (a) and (b); see Fig. 4 for (c).

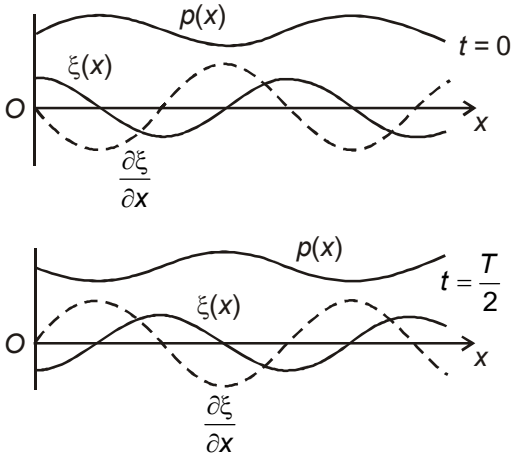


Fig. 4(a) and (b)

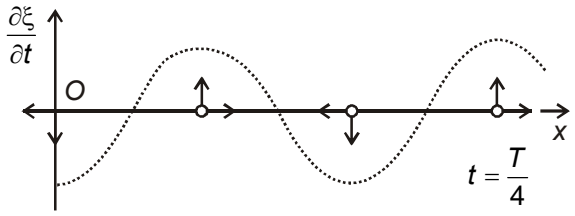


Fig. 4(c)

OPTICS

1. $\vec{e}' = \vec{e} - 2(\vec{e}\cdot\vec{n})\,\vec{n}$

2. (a) $f = \dfrac{\ell\beta}{(1-\beta^2)} = 10$ cm;

   (b) $f = \dfrac{\ell\beta_1\beta_2}{(\beta_2 - \beta_1)} = 2.5$ cm

3. $x = \dfrac{nf}{n+1}\left(1 - \sqrt{1 - \dfrac{(n+1)r^2}{(n-1)f^2}}\right)$, $r_{max} = f\sqrt{\dfrac{(n-1)}{(n+1)}}$.

4. $f = \dfrac{n_0 R}{2(n_1 - n_2)} = 35$ cm, where n_0 is refractive index of water.

5. $\Psi = \dfrac{\Psi'}{\sqrt{\eta}} = 0.6'$

6. (a) $\cos\theta = \left(k - \dfrac{\varphi}{2\pi}\right)\dfrac{\lambda}{d}$, k = 0, ±1, ±2, ...;

   (b) $\varphi = \dfrac{\pi}{2}$, $\dfrac{d}{\lambda} = k + \dfrac{1}{4}$, $k = 0, 1, 2, \ldots$

7. $\lambda = \dfrac{2\Delta x \Delta h}{\ell(\eta - 1)}$ = 0.6 μm

8. $\Delta x \approx \dfrac{\lambda}{2\Theta(n - n')} = 0.20$ mm

9. $n' = n + \dfrac{N\lambda}{\ell} = 1.000377$

10. $\Phi = \dfrac{2(n-1)(2k-1)\lambda}{d^2} = 2.4$ D, where k is the number of the bright ring.

11. $r = \sqrt{\dfrac{1}{2}\dfrac{(1+2k)\lambda R}{n_2}} = 1.3$ mm, where k = 5

12. $b' = \dfrac{b}{\eta^2} = 1.0$ m.

ATOMIC & NUCLEAR PHYSICS

1. (a) r_{min} = 0.59 pm;

   (b) $r_{min} = \left(\frac{2Ze^2}{T}\right)\left(1+\frac{m\alpha}{m\text{Li}}\right) = 0.034$ pm.

2. (a) $\rho_{min} = \left(\frac{Ze^2}{T}\right)\cot^2\left(\frac{\theta}{2}\right) = 0.23$ pm;

   (b) $r_{min} = \left[1+\text{cosec}\left(\frac{\theta}{2}\right)\right]\frac{Ze^2}{T} = 0.56$ pm;

3. $b = \frac{Rn\sin\left(\frac{\theta}{2}\right)}{\sqrt{1+n^2-2n\cos\left(\frac{\theta}{2}\right)}}$, where $n = \sqrt{1+\frac{U_0}{T}}$

4. (a) $\cos\left(\frac{\theta}{2}\right) = \frac{b}{(R+r)}$;

   (b) $dP = \frac{1}{2}\sin\theta\, d\theta$;

   (c) $P = \frac{1}{2}$

5. $t = \left(\frac{3mc^3}{2e^2\omega^2}\right)\ln\eta = 15$ ns.

6. $t \approx \frac{m^2c^3r^3}{4e^4} \approx 13$ ps

7. $r_n = \sqrt{\frac{n\hbar}{m\omega}}$, $E_n = n\hbar\omega$, where n = 1, 2, ..., $\omega = \sqrt{\frac{k}{m}}$

8. $\mu_n = \frac{ne\hbar}{2mc}$, $\frac{\mu_n}{M_n} = \frac{e}{2mc}$, $\mu_1 = \mu_B$.

9. n = 5

10. $v = 2\sqrt{\frac{\hbar R}{m}}$ = 3.1 × 10^6 m/s, where m is the mass of the electron.

11. (a) 0.285 pm, 2.53 keV, 0.65 nm;
    (b) 106 pm, 6.8 eV, 0.243 μm

12. $\tilde{\lambda} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2+\lambda_2^2}}$

13. $T = \left(\sqrt{2}-1\right)mc^2 = 0.21$ MeV

14. $V = \left(\frac{A}{A'}\right)\exp\left(\frac{-t\ln 2}{T}\right)$ = 6 L.

15. $t = -\left(\frac{T}{\ln 2}\right)\ln\left(1-\frac{A}{q}\right) = 9.5$ days.

16. (a) $N_2(t) = N_{10}\frac{\lambda_1}{\lambda_2-\lambda_1}\left(e^{-\lambda_1 t}-e^{-\lambda_2 t}\right)$;

(b) $t_m = \dfrac{\ln\left(\dfrac{\lambda_1}{\lambda_2}\right)}{\lambda_1 - \lambda_2}$.

17. (a) $N_2(t) = \lambda N_{10} t \exp(-\lambda t)$;

(b) $t_m = \dfrac{1}{\lambda}$

18. (a) $\eta = \dfrac{4mM}{(m+M)^2} = 0.89$;

(b) $\eta = \dfrac{2m}{(m+M)} = \dfrac{2}{3}$.

Here m and M are the masses of a neutron and a deuteron.

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Chemistry

Answer Key

1. (c)
2. (c)
3. (c)
4. (d)
5. (a)
6. (a)
7. (b)
8. (c)
9. (c)
10. (a)
11. (a)
12. (d)
13. (c)
14. (b)
15. (b)
16. (d)
17. (d)
18. (b)
19. (a, b)
20. A $\rightarrow$ (P, R), B $\rightarrow$ (Q, R), C $\rightarrow$ (Q, S), D $\rightarrow$ (Q, R)
21. A $\rightarrow$ (S), B $\rightarrow$ (P), C $\rightarrow$ (Q), D $\rightarrow$ (R)
22. (6)
23. (d)
24. (c)
25. (b)
26. (d)
27. (c)
28. (c)
29. (c)
30. (b)
31. (b)

32. (a, b, c, d)
33. A → (P), B → (P, Q, S), C → (Q, S), D → (Q, S)
34. (3)
35. (d)
36. (c)
37. (a, d)
38. (b)
39. (c)
40. (b)
41. (a)
42. (c)
43. (b)
44. (a)
45. (b)
46. (c)
47. (a)
48. (d)
49. (a)
50. (c)
51. (d)
52. (c)
53. (c)
54. (a)
55. (a, d)
56. (a, b, d)
57. A → (Q, S), B → (R), C → (P, T), D → (T)
58. (d)
59. (a)
60. (c)
61. (b)
62. (a)
63. (a)
64. (a)
65. (d)
66. (c)
67. (c)
68. (d)
69. (a)
70. (a)
71. (c, d)
72. (a, c, d)
73. a → (q), b → (p), c → (r), d → (s)

74. (6)
75. (a)
76. (b)
77. (c)
78. (b)
79. (b)
80. (a)
81. (d)
82. (c)
83. (a)
84. (a, b, c)
85. (b)
86. (c)
87. (d)
88. (a)
89. (c)
90. (c)
91. (c)
92. (b)
93. (c)
94. (a)
95. (b)
96. (a)
97. (b)
98. (d)
99. (a, c)
100. (4)
101. (b)
102. (c)
103. (a)
104. (c)
105. (d)
106. (d)
107. (c)
108. (b)
109. (a)
110. (d)
111. (c)

112. (d)
113. (d)
114. (b)
115. (a)
116. (d)
117. (b)
118. (b)
119. (c)
120. (a, b, c)
121. (25)
122. (7)
123. (c)
124. (d)
125. (d)
126. (d)
127. (c)
128. (c)
129. (b)
130. (d)
131. (b)
132. (a)
133. (c)
134. (b)
135. (a)
136. (c)
137. (b)
138. (a, b, c, d)
139. (a, b, c)
140. (a)
141. (a, b)
142. (a)
143. (d)
144. (b)
145. (a)
146. (d)
147. (c)
148. (c)

149. (c)
150. (b)
151. (a)
152. (d)
153. (d)
154. (d)
155. (c)
156. (c)
157. (b)
158. (b)
159. (b)
160. (c)
161. (c)
162. (b)
163. (b)
164. (d)
165. (b)
166. (a)
167. (d)
168. (a)
169. (b)
170. (b)
171. (a)
172. (a)
173. (a, b, c, d)
174. (a, c)
175. a $\rightarrow$ (p), b $\rightarrow$ (r), c $\rightarrow$ (q), d $\rightarrow$ (s)
176. a $\rightarrow$ (q), b $\rightarrow$ (q), c $\rightarrow$ (p), d $\rightarrow$ (r)
177. (4)
178. (4)
179. (a)
180. (b)
181. (b)
182. (a)
183. (b)
184. (c)
185. (a)
186. (c)
187. (b)

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ANSWERS

1.	(4)	2.	(3)	3.	(3)	4.	(3)	5.	(4)
6.	(4)	7.	(2)	8.	(1)	9.	(1, 3, 4)	10.	(1, 2, 3)
11.	(1, 3)	12.	(1, 2, 4)	13.	(1, 2, 3, 4)				
14.	A(r); B(s); C(q), D(p)								
15.	17	16.	20	17.	1	18.	(2)	19.	(3)
20.	(1)	21.	(1)	22.	(3)	23.	(4)	24.	(3)
25.	(1, 2, 3)	26.	(1, 4)	27.	(1, 2, 4)	28.	(1, 3)		
29.	A(p); B(q, r, s, t); C(q); D(r)								
30.	A(p); B(q); C(p); D(s)								
31.	A(s); B(r); C(s); D(q)								
32.	31	33.	720	34.	41	35.	47	36.	38
37.	(2)	38.	(2)	39.	(2)	40.	(2)	41.	(3)
42.	(1)	43.	(3)	44.	(3)	45.	(4)	46.	(3)
47.	(4)	48.	(2)	49.	(3)	50.	(2, 3, 4)	51.	(2, 4)
52.	(1, 3, 4)	53.	(2, 3)	54.	(1, 2, 4)	55.	(3, 4)		
56.	A(p, q, r, s, t); B(p, q, s, t); C(p, t); D(p, q, s, t)								
57.	A(s); B(p); C(r); D(q)								
58.	2	59.	5	60.	11	61.	17	62.	683
63.	2	64.	200	65.	4	66.	(2)	67.	(3)
68.	(3)	69.	(4)	70.	(1)	71.	(4)	72.	(4)
73.	(4)	74.	(3)	75.	(2, 3, 4)	76.	(1, 2, 4)		
77.	A(r); B(p); C(s); D(q)								
78.	300	79.	0	80.	(3)	81.	(1)	82.	(2)
83.	(1)	84.	(1)	85.	(1)	86.	(1, 4)	87.	12
88.	2	89.	(1)	90.	(2)	91.	(1)	92.	(1)
93.	(2)	94.	(3)	95.	(3)	96.	(2)	97.	(3)
98.	(3)	99.	(4)	100.	(2)	101.	(3)	102.	(3, 4)

103. (1, 2, 4) 104. (2, 3, 4)

105. A(r, s); B(p, r); C(q, r); D(r)

106. 1	107. 3	108. 43	109. 13	110. 1
111. 9	112. (4)	113. (3)	114. (2)	115. (2)
116. (3)	117. (4)	118. (3)	119. (4)	120. (1, 3, 4)
121. (1, 3)	122. (1)	123. (1, 2, 3, 4)		

124. A(p, r); B(p, q, r); C(r, s); d(p, q)

125. 9	126. 1	127. 13	128. 3	129. (2)
130. (1)	131. (1)	132. (1)	133. (2)	134. (2)
135. (2)	136. (1)	137. (1, 2, 4)	138. (1, 2, 3)	139. 9
140. (3)	141. (1)	142. (3)	143. (4)	144. (1)
145. (2)	146. (3)	147. (1, 2, 3, 4)	148. 4	149. 4

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